

Technical Description

SIGMA CONTROL 2

Process Map $\geq$ blow_2.5.2

7_7601_BLOW_PA_14E

General

1. In communication mode "send/receive" Sigma Control can monitor the master program. This takes place with "iSlaveMsgHeader" and "iSlaveMsgFooter" that have to be returned unchanged from master.

Example sequence for Profibus Master in Step 7 :

L DB (receive box master).DBW 94 Message header
 T DB (transmit box master).DBW 38
 L DB (receive box master).DBW 126 Message footer
 T DB (transmit box master).DBW 62

Monitoring can be activated in menu *"Communication / Com-Module / / Communication error: Timeout"* by entering a check mark at check box right of Timeout.
 If monitoring is not required the above sequence is not used.

2. For further information on signal availability and use ask your local KAESER subsidiary or KAESER Headquarter Service Support Engineering, phone +49 9561 640-7979

Sigma Control 2 Profibus

1. The data required for planning a Profibus project are in the equipment specification file "KAES0CEC.gsd".
2. The Sigma Control is a Profibus DPV0 slave with 64 bytes of input data and 128 bytes of output data. Depending on the number of Sigma Control units used the Profibus DP master must have a correspondingly large enough address space.

Sigma Control 2 Profinet

1. The data required for planning a Profinet project are in the equipment specification file "GSDML-V2.25-Kaeser-SC2-20120203.xml".

Sigma Control 2 Modbus**1. Blower ==>>>> Master Controller**

Read Input Registers with modbus function 04 "Read Input Registers" or read Holding Registers with modbus function 03 "Read Holding Registers".
 Modbus Protocol address: Address of the data at the bus (Base 0).
 Modbus Register: Register address of the data = PLC-address (Base 1).
 3xxxx 3 = Identification of Input Register, xxxx = Register address
 4xxxx 4 = Identification of Holding Register, xxxx = Register address

2. Master Controller ==>>>> Blower

Write registers with function 16 "Write Multiple Registers" or 06 "Write Single Register".
 Read registers with function 03 "Read Holding Registers". Never written registers contain "0".
 Modbus Protocol address: Address of the data at the bus (Base 0).
 Modbus Register: Register address of the data = PLC-address (Base 1).
 4xxxx 4 = Identification of Holding Register, xxxx = Register address

Sigma Control 2 Devicenet

1. The data required for planning a DeviceNet project are in the equipment specification file "324-8172-EDS_ABCC_DEV_V_2_3_1.eds".
2. 8 Bit register: Low bytes are shown at low address.

Sigma Control 2 EtherNet/IP

1. The data required for planning an EtherNet/IP project are in the equipment specification file "SC2_EtherNetIP.eds".
2. 8 Bit register: Low bytes are shown at low address.

SAM Send/Receive

1. Data for blower 1 to 16 can be found in SAM data block 64 to 79.

SIGMA CONTROL 2 Process Map

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Table 1 Process

1. Blower ==>>>> Master Controller

Table 1: Process data

* -32768 = Value invalid or not used

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 2

JSON-RPC	Sigma Control 2							SAM		Description	Unit	min	max	Format	
	Profibus/-net		Modbus **				DeviceNet	Send/Receive							
			Input		Holding		EtherNet/IP								
BDSnd	S7 DBW	S5 DW	Protocol address	Input Register	Protocol address	Holding Register	Register MSB/LSB	S5 DW	S7 DBW						
	0	0	21	277	30022	40278	1/0	32	64	Alarm xAM16 - xAM1	Table 1.1	binary			BOOL
	2	1	22	278	30023	40279	3/2	33	66	Alarm xAM32 - xAM17	Table 1.1	binary			BOOL
	4	2	23	279	30024	40280	5/4	34	68	Alarm xAM48 - xAM33	Table 1.1	binary			BOOL
	6	3	24	280	30025	40281	7/6	35	70	Alarm xAM64 - xAM49	Table 1.1	binary			BOOL
	8	4	25	281	30026	40282	9/8	36	72	Alarm xAM80 - xAM65	Table 1.1	binary			BOOL
	10	5	26	282	30027	40283	11/10	37	74	Warning xWM16 - xWM1	Table 1.2	binary			BOOL
	12	6	27	283	30028	40284	13/12	38	76	Warning xWM32 - xWM17	Table 1.2	binary			BOOL
	14	7	28	284	30029	40285	15/14	39	78	Warning xWM48 - xWM33	Table 1.2	binary			BOOL
	16	8	29	285	30030	40286	17/16	40	80	Warning xWM64 - xWM49	Table 1.2	binary			BOOL
	18	9	30	286	30031	40287	19/18	41	82	Warning xWM80 - xWM65	Table 1.2	binary			BOOL
	20	10	31	287	30032	40288	21/20	42	84	Operational message xOM16 - xOM1	Table 1.3	binary			BOOL
	22	11	32	288	30033	40289	23/22	43	86	Operational message xOM32 - xOM17	Table 1.3	binary			BOOL
	24	12	33	289	30034	40290	25/24	44	88	Operational message xOM40 - xOM33	Table 1.3	binary			BOOL
	26	13	34	290	30035	40291	27/26	45	90	Binary signal xBS15 - xBS0	Table 1.4	binary			BOOL
	28	14	35	291	30036	40292	29/28	46	92	Binary signal xBS31 - xBS16	Table 1.4	binary			BOOL
	30	15	36	292	30037	40293	31/30	47	94	Binary signal xBS47 - xBS32	Table 1.4	binary			BOOL
	32	16	37	293	30038	40294	33/32	48	96	Binary signal xBS63 - xBS48	Table 1.4	binary			BOOL
iDV0	34	17	38	294	30039	40295	35/34	49	98	Software version 3rd digit / 4th digit					BYTE/BYTE
iDV1	36	18	39	295	30040	40296	37/36	50	100	Software version 1st digit / 2nd digit					BYTE/BYTE
iDV2	38	19	40	296	30041	40297	39/38	51	102	free					
iDV3	40	20	41	297	30042	40298	41/40	52	104	Pressure actual value (Pressure: p2, Vacuum: p1)		mbar	-2000*	2000	INT
iDV4	42	21	42	298	30043	40299	43/42	53	106	Internal pressure (Pressure: p1, Vacuum: p2)		mbar	-2000*	2000	INT
iDV5	44	22	43	299	30044	40300	45/44	54	108	Airend discharge temperature t2		0.1 °C	-500*	2500	INT
iDV6	46	23	44	300	30045	40301	47/46	55	110	Temperature sound enclosure (t_S)		0.1 °C	-500*	2500	INT
iDV7	48	24	45	301	30046	40302	49/48	56	112	Inlet temperature t1		0.1 °C	-500*	2500	INT
iDV8	50	25	46	302	30047	40303	51/50	57	114	Oil temperature t3		0.1 °C	-500*	2500	INT
iDV9	52	26	47	303	30048	40304	53/52	58	116	Block differential pressure		mbar	-2000*	2000	INT
iDV10	54	27	48	304	30049	40305	55/54	59	118	Block temperature difference		0.1 °C	-500*	2500	INT
iDV11	56	28	49	305	30050	40306	57/56	60	120	Main motor winding temperature (t_M)		0.1 °C	-500*	2500	INT
iDV12	58	29	50	306	30051	40307	59/58	61	122	free					
iDV13	60	30	51	307	30052	40308	61/60	62	124	free					
iDV14	62	31	52	308	30053	40309	63/62	63	126	Current drive motor starts/h		1			INT

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Table 1 Process

1. Blower ==>>>> Master Controller

Table 1: Process data

* -32768 = Value invalid or not used

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 2

JSON-RPC	Sigma Control 2							SAM		Description	Unit	min	max	Format
	Profibus/-net		Modbus **				DeviceNet EtherNet/IP Register MSB/LSB	Send/Receive						
	S7 DBW	S5 DW	Input Protocol address	Input Register	Holding Protocol address	Holding Register		S5 DW	S7 DBW					
iDV15	64	32	53	309	30054	40310	65/64	64	128	Total hours run x 1000	h			INT
iDV16	66	33	54	310	30055	40311	67/66	65	130	Total hours run x 1	h			INT
iDV17	68	34	55	311	30056	40312	69/68	66	132	Hours on load x 1000	h			INT
iDV18	70	35	56	312	30057	40313	71/70	67	134	Hours on load x 1	h			INT
iDV19	72	36	57	313	30058	40314	73/72	68	136	Current motor frequency	Hz			INT
uiDV20	74	37	58	314	30059	40315	75/74	69	138	Frequency converter output current	0.01A			UINT
iDV21	76	38	59	315	30060	40316	77/76	70	140	Frequency converter direct current link voltage	V			INT
iDV22	78	39	60	316	30061	40317	79/78	71	142	Frequency converter torque	0.1 Nm			INT
iDV23	80	40	61	317	30062	40318	81/80	72	144	free				
iDV24	82	41	62	318	30063	40319	83/82	73	146	free				
iDV25	84	42	63	319	30064	40320	85/84	74	148	free				
iDV26	86	43	64	320	30065	40321	87/86	75	150	Analogue input All 1.02	0.001 mA	-2000*	22000	INT
iDV27	88	44	65	321	30066	40322	89/88	76	152	free				
iDV28	90	45	66	322	30067	40323	91/90	77	154	free				
iDV29	92	46	67	323	30068	40324	93/92	78	156	free				
iSlaveMsgHeader	94	47	68	324	30069	40325	95/94	79	158	Message header slave				WORD
iDV30	96	48	69	325	30070	40326	97/96	80	160	free				
iDV31	98	49	70	326	30071	40327	99/98	81	162	Remaining service life lubricating oil	h			INT
iDV32	100	50	71	327	30072	40328	101/100	82	164	free				
iDV33	102	51	72	328	30073	40329	103/102	83	166	Remaining service life inlet filter	h			INT
iDV34	104	52	73	329	30074	40330	105/104	84	168	Remaining service life filter mat	h			INT
iDV35	106	53	74	330	30075	40331	107/106	85	170	Remaining service life V-belts	h			INT
uiDV36	108	54	75	331	30076	40332	109/108	86	172	Remaining service life motor bearings (positive value)	h			UINT
iDV37	110	55	76	332	30077	40333	111/110	87	174	Motor bearing service life exceeded (negative value)	h			INT
iDV38	112	56	77	333	30078	40334	113/112	88	176	Remaining time till relubrication of motor bearings	h			INT
iDV39	114	57	78	334	30079	40335	115/114	89	178	free				
iDV40	116	58	79	335	30080	40336	117/116	90	180	Current motor speed	1/min			INT
iDV41	118	59	80	336	30081	40337	119/118	91	182	Blower model / Slave address				BYTE/BYTE
iDV42	120	60	81	337	30082	40338	121/120	92	184	free				
iDV43	122	61	82	338	30083	40339	123/122	93	186	Duty cycle	1%			INT
iDV44	124	62	83	339	30084	40340	125/124	94	188	Current block speed	1/min			INT
iSlaveMsgFooter	126	63	84	340	30085	40341	127/126	95	190	Message footer slave				WORD

1. Blower ==>>>> Master Controller

Tabelle 1: Diagnostic data

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 2

* since manuf. date MFGDT: 2010/11

JSON-RPC		Sigma Control 2							SAM		Description	Example		
		Profibus		Modbus **				DeviceNet EtherNet/IP	Send/Receive			HEX	Value	Result
BDSnd	S7 DBB	S5 DW	Input address	Input Register	Protocol address	Holding Register	Byte	Register MSB/LSB	S5 DW	S7 DBB				
%	0	DL0	%	%	%	%		%	DL96	192	Profibus: System status 1			
%	1	DR0	%	%	%	%		%	DR96	193	Profibus: System status 2			
%	2	DL1	%	%	%	%		%	DL97	194	Profibus: System status 3			
%	3	DR1	%	%	%	%		%	DR97	195	Profibus: DP master address	01	1	
%	4	DL2	%	%	%	%		%	DL98	196	Profibus: Manufacturer code: PNO number	0C		Kaeser PNO Nr.: 0CEC
%	5	DR2	%	%	%	%		%	DR98	197	Profibus: Manufacturer code: PNO number	EC		
%	6	DL3	%	%	%	%		%	DL99	198	Profibus: Header-Byte specific diagnostic data			
%	7	DR3	1	30002	257	40258	MSB	%	DR99	199	SC2 MCS Kaeser SN: 5th digit, 6th digit *	62	62	SC2 MCS Kaeser SN: 003062
%	8	DL4	1	30002	257	40258	LSB	%	DL100	200	SC2 MCS Kaeser SN: 3rd digit, 4th digit *	30	30	
%	9	DR4	2	30003	258	40259	MSB	%	DR100	201	SC2 MCS Kaeser SN: 1st digit, 2nd digit *	00	00	
%	10	DL5	2	30003	258	40259	LSB	%	DL101	202	reserved	00		
%	11	DR5	8	30009	264	40265	MSB	%	DR101	203	SC2 MCS Kaeser PN: space (ASCII)	20	SP	SC2 MCS Kaeser PN: 7.7601P0
%	12	DL6	8	30009	264	40265	LSB	%	DL102	204	SC2 MCS Kaeser PN: space (ASCII)	20	SP	
%	13	DR6	9	30010	265	40266	MSB	%	DR102	205	SC2 MCS Kaeser PN: space (ASCII)	20	SP	
%	14	DL7	9	30010	265	40266	LSB	%	DL103	206	SC2 MCS Kaeser PN: space (ASCII)	20	SP	
%	15	DR7	10	30011	266	40267	MSB	%	DR103	207	SC2 MCS Kaeser PN: space (ASCII)	20	SP	
%	16	DL8	10	30011	266	40267	LSB	%	DL104	208	SC2 MCS Kaeser PN: space (ASCII)	20	SP	
%	17	DR8	11	30012	267	40268	MSB	%	DR104	209	SC2 MCS Kaeser PN: space (ASCII)	20	SP	
%	18	DL9	11	30012	267	40268	LSB	%	DL105	210	SC2 MCS Kaeser PN: space (ASCII)	20	SP	
%	19	DR9	4	30005	260	40261	MSB	%	DR105	211	SC2 MCS Kaeser PN: 8th digit (ASCII)	30	0	
%	20	DL10	4	30005	260	40261	LSB	%	DL106	212	SC2 MCS Kaeser PN: 7th digit (ASCII)	50	P	
%	21	DR10	5	30006	261	40262	MSB	%	DR106	213	SC2 MCS Kaeser PN: 6th digit (ASCII)	31	1	
%	22	DL11	5	30006	261	40262	LSB	%	DL107	214	SC2 MCS Kaeser PN: 5th digit (ASCII)	30	0	
%	23	DR11	6	30007	262	40263	MSB	%	DR107	215	SC2 MCS Kaeser PN: 4th digit (ASCII)	36	6	
%	24	DL12	6	30007	262	40263	LSB	%	DL108	216	SC2 MCS Kaeser PN: 3rd digit (ASCII)	37	7	
%	25	DR12	7	30008	263	40264	MSB	%	DR108	217	SC2 MCS Kaeser PN: 2nd digit (ASCII)	2E	.	
%	26	DL13	7	30008	263	40264	LSB	%	DL109	218	SC2 MCS Kaeser PN: 1st digit (ASCII)	37	7	

SIGMA CONTROL 2 Process Map

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Table 1 Diag

1. Blower ==>>>> Master Controller

Tabelle 1: Diagnostic data

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 2

* since manuf. date MFGDT: 2010/11

Sigma Control 2									SAM		Description	Example		
JSON-RPC	Profibus		Modbus **					DeviceNet EtherNet/IP	Send/Receive			HEX	Value	Result
BDSnd	S7 DBB	S5 DW	Input Protocol address	Input Register	Protocol address	Holding Register	Byte	Register MSB/LSB	S5 DW	S7 DBB				
%	27	DR13	12	30013	268	40269	MSB	%	DR109	219	reserved	00		
%	28	DL14	12	30013	268	40269	LSB	%	DL110	220	reserved	00		
%	29	DR14	13	30014	269	40270	MSB	%	DR110	221	reserved	00		
%	30	DL15	13	30014	269	40270	LSB	%	DL111	222	reserved	00		
%	%	%	14	30015	270	40271	MSB	%	%	%	reserved	00		
%	%	%	14	30015	270	40271	LSB	%	%	%	reserved	00		
%	%	%	15	30016	271	40272	MSB	%	%	%	reserved	00		
%	%	%	15	30016	271	40272	LSB	%	%	%	reserved	00		
%	%	%	16	30017	272	40273	MSB	%	%	%	reserved	00		
%	%	%	16	30017	272	40273	LSB	%	%	%	reserved	00		
%	%	%	17	30018	273	40274	MSB	%	%	%	reserved	00		
%	%	%	17	30018	273	40274	LSB	%	%	%	reserved	00		
%	%	%	18	30019	274	40275	MSB	%	%	%	Modbus: Controller identifier (ASCII)	20	SP	SC2
%	%	%	18	30019	274	40275	LSB	%	%	%	Modbus: Controller identifier (ASCII)	53	S	
%	%	%	19	30020	275	40276	MSB	%	%	%	Modbus: Controller identifier (ASCII)	43	C	
%	%	%	19	30020	275	40276	LSB	%	%	%	Modbus: Controller identifier (ASCII)	32	2	

SIGMA CONTROL 2 Process Map

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Table 1.1

1. Blower ==>>> Master Controller

Table 1.1 : Alarms

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 2

* Summary of SC2 messages

JSON-RPC	Sigma Control 2								SAM		Message		Description / possible cause	
	Profibus-net		Modbus**				DeviceNet EtherNet/IP	Send/Receive						
	BDSnd	S7 DBX	S5 DW	Input address	Input Register	Holding address		Holding Register	S5 Bit	Register	Bit	S5 DW		S7 DBX
xAM1	1.0	0	21	30022	277	40278	0	0	0	32	65.0		free	
xAM2	1.1	0	21	30022	277	40278	1	0	1	32	65.1	0016	t_M‡	Drive motor temperature is too high
xAM3	1.2	0	21	30022	277	40278	2	0	2	32	65.2		free	
xAM4	1.3	0	21	30022	277	40278	3	0	3	32	65.3	0001	Emergency stop	EMERGENCY STOP control device actuated
xAM5	1.4	0	21	30022	277	40278	4	0	4	32	65.4	0008	Oil level side A	Lubricant level at the blower block drive side too low
xAM6	1.5	0	21	30022	277	40278	5	0	5	32	65.5	0020	Oil level side B	Lubricant level at the blower block control side too low
xAM7	1.6	0	21	30022	277	40278	6	0	6	32	65.6	0006	Phase error blower motor	Electric feed has incorrect phase sequence
xAM8	1.7	0	21	30022	277	40278	7	0	7	32	65.7	0024	Diagnostics group alarm	A group alarm occurred.
xAM9	0.0	0	21	30022	277	40278	8	1	0	32	64.0	*	Sigma Control T‡	Permissible internal temperature for SIGMA CONTROL 2 MCS or IOM exceeded
xAM10	0.1	0	21	30022	277	40278	9	1	1	32	64.1	0010	p2 ‡	Discharge pressure too high
xAM11	0.2	0	21	30022	277	40278	10	1	2	32	64.2		free	
xAM12	0.3	0	21	30022	277	40278	11	1	3	32	64.3		free	
xAM13	0.4	0	21	30022	277	40278	12	1	4	32	64.4	0002	Overload blower motor	Current draw of the blower motor too high
xAM14	0.5	0	21	30022	277	40278	13	1	5	32	64.5	0022	Aftercooler	Current draw of the aftercooler fan motor too high
xAM15	0.6	0	21	30022	277	40278	14	1	6	32	64.6	0012	t2 ‡	Permissible airend discharge temperature (ADT) exceeded
xAM16	0.7	0	21	30022	277	40278	15	1	7	32	64.7		free	

SIGMA CONTROL 2 Process Map

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Table 1.1

1. Blower ==>>>> Master Controller

Table 1.1 : Alarms

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 2

* Summary of SC2 messages

JSON-RPC	Sigma Control 2										SAM		Message		Description / possible cause
	Profibus/-net		Modbus**				Modbus	DeviceNet		Send/Receive					
	BDSnd	S7 DBX	S5 DW	Input address	Input Register	Holding address		Holding Register	S5 Bit	Register	Bit	S5 DW	S7 DBX	ID	
xAM17	3.0	1	22	30023	278	40279	0	2	0	33	67.0		free		
xAM18	3.1	1	22	30023	278	40279	1	2	1	33	67.1	0003	Fan motor	Current draw of the fan motor too high	
xAM19	3.2	1	22	30023	278	40279	2	2	2	33	67.2	0009	p1 ‡	Inlet pressure too low. Inlet filter clogged.	
xAM20	3.3	1	22	30023	278	40279	3	2	3	33	67.3	0013	t_S ‡	Sound enclosure temperature too high	
xAM21	3.4	1	22	30023	278	40279	4	2	4	33	67.4		free		
xAM22	3.5	1	22	30023	278	40279	5	2	5	33	67.5	0011	dp ‡	Differential pressure too high	
xAM23	3.6	1	22	30023	278	40279	6	2	6	33	67.6	0015	t1 ‡	Inlet temperature too high	
xAM24	3.7	1	22	30023	278	40279	7	2	7	33	67.7	0027	t1 ‡	Inlet temperature too low	
xAM25	2.0	1	22	30023	278	40279	8	3	0	33	66.0	*	IOM - Error	Group alarm IOM. Error in IOM itself.	
xAM26	2.1	1	22	30023	278	40279	9	3	1	33	66.1		free		
xAM27	2.2	1	22	30023	278	40279	10	3	2	33	66.2	0018	Filter blocked	Inlet filter clogged	
xAM28	2.3	1	22	30023	278	40279	11	3	3	33	66.3	0025	t3 ‡	Oil temperature is very high	
xAM29	2.4	1	22	30023	278	40279	12	3	4	33	66.4		free		
xAM30	2.5	1	22	30023	278	40279	13	3	5	33	66.5		free		
xAM31	2.6	1	22	30023	278	40279	14	3	6	33	66.6	0017	dT ‡	t2 / t1 temperature difference too high	
xAM32	2.7	1	22	30023	278	40279	15	3	7	33	66.7	0026	π=p2_a/p1_a ‡	Pressure ratio across the block is too high	

SIGMA CONTROL 2 Process Map

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Table 1.1

1. Blower ==>>> Master Controller

Table 1.1 : Alarms

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 2

* Summary of SC2 messages

JSON-RPC	Sigma Control 2										SAM		Message		Description / possible cause
	Profibus/-net		Modbus**				Modbus	DeviceNet		Send/Receive					
			Input		Holding			EtherNet/IP							
BDSnd	S7 DBX	S5 DW	Protocol address	Input Register	Protocol address	Holding Register	S5 Bit	Register	Bit	S5 DW	S7 DBX	ID	Text (SC2 Display, except *)		
xAM33	5.0	2	23	30024	279	40280	0	4	0	34	69.0		free		
xAM34	5.1	2	23	30024	279	40280	1	4	1	34	69.1		free		
xAM35	5.2	2	23	30024	279	40280	2	4	2	34	69.2	0035	dp ‡	Pressure difference across the block too high	
xAM36	5.3	2	23	30024	279	40280	3	4	3	34	69.3		free		
xAM37	5.4	2	23	30024	279	40280	4	4	4	34	69.4		free		
xAM38	5.5	2	23	30024	279	40280	5	4	5	34	69.5	0038	Drag indicator p1 ‡	Inlet pressure too low. Inlet filter clogged.	
xAM39	5.6	2	23	30024	279	40280	6	4	6	34	69.6	0039	Drag indicator p2 ‡	Discharge pressure too high	
xAM40	5.7	2	23	30024	279	40280	7	4	7	34	69.7	0004	Mains contactor feedback signal	Mains contactor does not close or does not open	
xAM41	4.0	2	23	30024	279	40280	8	5	0	34	68.0	0040	Overvoltage suppressor	Overvoltage at power supply.	
xAM42	4.1	2	23	30024	279	40280	9	5	1	34	68.1	0140	AFxM-valve actuator error	AFxM-valve actuator fault	
xAM43	4.2	2	23	30024	279	40280	10	5	2	34	68.2		free		
xAM44	4.3	2	23	30024	279	40280	11	5	3	34	68.3	0005 0104	Motor stop or block speed	Fault in drive system. Block speed is outside the permissible range.	
xAM45	4.4	2	23	30024	279	40280	12	5	4	34	68.4	0007	Vibration alarm	Vibration too high	
xAM46	4.5	2	23	30024	279	40280	13	5	5	34	68.5	0019	Error FC ramp up	Internal fault frequency converter	
xAM47	4.6	2	23	30024	279	40280	14	5	6	34	68.6	0023	USS bus communication error	RS485-FC: USS bus communication error	
xAM48	4.7	2	23	30024	279	40280	15	5	7	34	68.7	*	Error Power switching FC contact	Group alarm of the power switching FC contact	

SIGMA CONTROL 2 Process Map

≥ blow_2.5.2

Table 1.1

1. Blower ==>>>> Master Controller

Table 1.1 : Alarms

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 2

* Summary of SC2 messages

JSON-RPC	Sigma Control 2										SAM		Message		Description / possible cause
	Profibus/-net		Modbus**				DeviceNet EtherNet/IP	Send/Receive							
			Input		Holding			Bit	S5	S7					
BDSnd	S7 DBX	S5 DW	Protocol address	Input Register	Protocol address	Holding Register	S5 Bit	Register	Bit	S5 DW	S7 DBX	ID	Text (SC2 Display, except *)		
xAM49	7.0	3	24	30025	280	40281	0	6	0	35	71.0		free		
xAM50	7.1	3	24	30025	280	40281	1	6	1	35	71.1	0050	Relubrication device	Relubrication device has been reporting an error for a long time	
xAM51	7.2	3	24	30025	280	40281	2	6	2	35	71.2		free		
xAM52	7.3	3	24	30025	280	40281	3	6	3	35	71.3		free		
xAM53	7.4	3	24	30025	280	40281	4	6	4	35	71.4	0028	Error USS handler	FC: Blower USS bus fault	
xAM54	7.5	3	24	30025	280	40281	5	6	5	35	71.5	0029	PKW requests failed	FC: USS Communication fault	
xAM55	7.6	3	24	30025	280	40281	6	6	6	35	71.6	0030	Error FC ramp up	FC: Error while activating the FC	
xAM56	7.7	3	24	30025	280	40281	7	6	7	35	71.7	0031	Motor: n min not reached	FC: Minimum speed not reached	
xAM57	6.0	3	24	30025	280	40281	8	7	0	35	70.0	0032	Heavy load monitoring	FC: Heavy load monitoring tripped	
xAM58	6.1	3	24	30025	280	40281	9	7	1	35	70.1	0120	p1 sensor error	Fault pressure sensor inlet side blower block	
xAM59	6.2	3	24	30025	280	40281	10	7	2	35	70.2	0121	p2 sensor error	Fault pressure sensor discharge side blower block	
xAM60	6.3	3	24	30025	280	40281	11	7	3	35	70.3	0122	Error external analogue input	Fault at the external analogue input	
xAM61	6.4	3	24	30025	280	40281	12	7	4	35	70.4	0123	t2 sensor error	Fault sensor PT100 discharge side blower block	
xAM62	6.5	3	24	30025	280	40281	13	7	5	35	70.5	0124	t_S sensor error	Fault sensor PT100 sound enclosure	
xAM63	6.6	3	24	30025	280	40281	14	7	6	35	70.6	0125	t1 sensor error	Fault sensor PT100 inlet side blower block	
xAM64	6.7	3	24	30025	280	40281	15	7	7	35	70.7	0126	t_M sensor error	Fault sensor PT100 blower motor	

SIGMA CONTROL 2 Process Map

≥ blow_2.5.2

Table 1.1

1. Blower ==>>> Master Controller

Table 1.1 : Alarms

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 2

* Summary of SC2 messages

JSON-RPC	Profibus/-net		Sigma Control 2						DeviceNet EtherNet/IP		SAM		Message		Description / possible cause
			Modbus**				Send/Receive								
	BDSnd	S7 DBX	S5 DW	Input address	Input Register	Holding address	Holding Register	Modbus S5 Bit	Register	Bit	S5 DW	S7 DBX	ID	Text (SC2 Display, except *)	
xAM65	9.0	4	25	30026	281	40282	0	8	0	36	73.0	0127	t3 sensor error	Fault sensor PT100 oil temperature	
xAM66	9.1	4	25	30026	281	40282	1	8	1	36	73.1		free		
xAM67	9.2	4	25	30026	281	40282	2	8	2	36	73.2		free		
xAM68	9.3	4	25	30026	281	40282	3	8	3	36	73.3		free		
xAM69	9.4	4	25	30026	281	40282	4	8	4	36	73.4		free		
xAM70	9.5	4	25	30026	281	40282	5	8	5	36	73.5	0080	Vibration monitoring VSE ‡	Vibrations at block or motor too high	
xAM71	9.6	4	25	30026	281	40282	6	8	6	36	73.6	0097	T-Switch T3		
xAM72	9.7	4	25	30026	281	40282	7	8	7	36	73.7	0098	T-Switch T_S		
xAM73	8.0	4	25	30026	281	40282	8	9	0	36	72.0	0021	External message 1	External binary message 1	
xAM74	8.1	4	25	30026	281	40282	9	9	1	36	72.1	0074	External message 2	External binary message 2	
xAM75	8.2	4	25	30026	281	40282	10	9	2	36	72.2	0075	External message 3	External binary message 3	
xAM76	8.3	4	25	30026	281	40282	11	9	3	36	72.3	0092	p-Switch p1	Adjustable p1-pressure switch	
xAM77	8.4	4	25	30026	281	40282	12	9	4	36	72.4	0093	p-Switch p2	Adjustable p2-pressure switch	
xAM78	8.5	4	25	30026	281	40282	13	9	5	36	72.5	0094	T-Switch T1	Adjustable T1-temperature switch	
xAM79	8.6	4	25	30026	281	40282	14	9	6	36	72.6	0095	T-Switch T2	Adjustable T2-temperature switch	
xAM80	8.7	4	25	30026	281	40282	15	9	7	36	72.7	0096	Block speed switch	Adjustable block speed switch	

1. Blower ==>>> Master Controller

Table 1.2 : Warning Messages

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 2

* Summary of SC2 messages

JSON-RPC BDSnd	Profibus-net		Sigma Control 2 Modbus**				Modbus S5 Bit	DeviceNet EtherNet/IP		SAM Send/Receive		Message		Description / possible cause
	S7 DBX	S5 DW	Input Protocol address	Input Register	Holding Protocol address	Holding Register		Register	Bit	S5 DW	S7 DBX			
xWM1	11.0	5	26	30027	282	40283	0	10	0	37	75.0	0001	Equipment number incomplete	The blower equipment number consists of less than 7 characters
xWM2	11.1	5	26	30027	282	40283	1	10	1	37	75.1	*	Collective warning	A group warning message occurred.
xWM3	11.2	5	26	30027	282	40283	2	10	2	37	75.2		free	
xWM4	11.3	5	26	30027	282	40283	3	10	3	37	75.3	0010	Oil level side A	Lubricant level at the blower block drive side too low
xWM5	11.4	5	26	30027	282	40283	4	10	4	37	75.4	0011	Oil level side B	Lubricant level at the blower block control side too low
xWM6	11.5	5	26	30027	282	40283	5	10	5	37	75.5	0005	t1 ↑	Inlet temperature too high
xWM7	11.6	5	26	30027	282	40283	6	10	6	37	75.6	0019	t1 ↓	Inlet temperature too low
xWM8	11.7	5	26	30027	282	40283	7	10	7	37	75.7	0006	t2 ↑	Discharge temperature too high
xWM9	10.0	5	26	30027	282	40283	8	11	0	37	74.0	0009	t_S ↑	Sound enclosure temperature too high
xWM10	10.1	5	26	30027	282	40283	9	11	1	37	74.1	0004	t_S ↓	Ambient temperature too low
xWM11	10.2	5	26	30027	282	40283	10	11	2	37	74.2	0003	dp ↑	Differential pressure too high
xWM12	10.3	5	26	30027	282	40283	11	11	3	37	74.3	0013	dp ↓	Differential pressure too low
xWM13	10.4	5	26	30027	282	40283	12	11	4	37	74.4	0012	Filter blocked	Inlet filter clogged.
xWM14	10.5	5	26	30027	282	40283	13	11	5	37	74.5	0017	t3 ↑	Oil temperature too high
xWM15	10.6	5	26	30027	282	40283	14	11	6	37	74.6	0015	Com-Module communication error	The bus link from the Com-Modul interface is interrupted
xWM16	10.7	5	26	30027	282	40283	15	11	7	37	74.7	0097	SAM 4.0 communication error	The bus link from the Com-Modul interface is interrupted (Togglebit)

1. Blower ==>>>> Master Controller

Table 1.2 : Warning Messages

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 2

* Summary of SC2 messages

JSON-RPC BDSnd	Profibus-net		Sigma Control 2 Modbus**				Modbus S5 Bit	DeviceNet EtherNet/IP		SAM Send/Receive		Message		Description / possible cause
	S7 DBX	S5 DW	Input Protocol address	Input Register	Holding Protocol address	Holding Register		Register	Bit	S5 DW	S7 DBX	ID	Text (SC2 Display, except *)	
xWM17	13.0	6	27	30028	283	40284	0	12	0	38	77.0	0007	dt ↑	t2 / t1 temperature difference too high
xWM18	13.1	6	27	30028	283	40284	1	12	1	38	77.1	0020	t_M ↑	Blower motor temperature too high
xWM19	13.2	6	27	30028	283	40284	2	12	2	38	77.2	0021	p1 ↓	Inlet pressure too low
xWM20	13.3	6	27	30028	283	40284	3	12	3	38	77.3	0002	p2 ↑	Final pressure too high
xWM21	13.4	6	27	30028	283	40284	4	12	4	38	77.4	0016	t3 ‡	Oil temperature too low
xWM22	13.5	6	27	30028	283	40284	5	12	5	38	77.5	0018	π=p2_a/p1_a ↑	Pressure ratio across the block is too high
xWM23	13.6	6	27	30028	283	40284	6	12	6	38	77.6		free	
xWM24	13.7	6	27	30028	283	40284	7	12	7	38	77.7	0008	Exchange mains contactor	Maximum number of switching cycles for the mains contactor reached
xWM25	12.0	6	27	30028	283	40284	8	13	0	38	76.0	0208	Check control cabinet fan	Motor bearing of control cabinet fan motor: Maintenance interval has elapsed
xWM26	12.1	6	27	30028	283	40284	9	13	1	38	76.1	0200	Change lube h ↑	Lubricating oil: Maintenance interval has elapsed
xWM27	12.2	6	27	30028	283	40284	10	13	2	38	76.2		free	
xWM28	12.3	6	27	30028	283	40284	11	13	3	38	76.3	0202	Change filter h ↑	Inlet filter: Maintenance interval has elapsed
xWM29	12.4	6	27	30028	283	40284	12	13	4	38	76.4	0203	Change filter mat h ↑	Air filter mat: Maintenance interval has elapsed
xWM30	12.5	6	27	30028	283	40284	13	13	5	38	76.5	0201	Change belt h ↑	V-belt: Maintenance interval has elapsed
xWM31	12.6	6	27	30028	283	40284	14	13	6	38	76.6	0206	Change motor bearings h ↑	Motor bearing of drive motor: Maintenance interval has elapsed
xWM32	12.7	6	27	30028	283	40284	15	13	7	38	76.7	0205	Check motor bearings h ↑	Motor bearing of drive motor inspection

1. Blower ==>>>> Master Controller

Table 1.2 : Warning Messages

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 2

* Summary of SC2 messages

Sigma Control 2												SAM		Message		Description / possible cause
JSON-RPC	Profibus-net		Modbus**				Modbus	DeviceNet		Send/Receive						
	BDSnd	S7 DBX	S5 DW	Input address	Input Register	Holding address		Holding Register	S5 Bit	DeviceNet EtherNet/IP	S5 DW	S7 DBX				
				Protocol address	Register	Protocol address	Register	Bit	Register	Bit			ID	Text (SC2 Display, except *)		
xWM33	15.0	7	28	30029	284	40285	0	14	0	39	79.0	0207	Check motor bearings encl. fan h ↑	Motor bearing of sound enclosure fan motor: Maintenance interval has elapsed		
xWM34	15.1	7	28	30029	284	40285	1	14	1	39	79.1	0209	Check FC fan h ↑	Motor bearing of frequency converter fan motor: Maintenance interval has elapsed		
xWM35	15.2	7	28	30029	284	40285	2	14	2	39	79.2	0211	Filter change transmission venting h ↑	Filter transmission venting: Maintetance interval has elapsed		
xWM36	15.3	7	28	30029	284	40285	3	14	3	39	79.3	0022	Motor-Starts ↑	Max. switching frequency/h exceeded		
xWM37	15.4	7	28	30029	284	40285	4	14	4	39	79.4		free			
xWM38	15.5	7	28	30029	284	40285	5	14	5	39	79.5	0038	Drag indicator p1 ‡	Inlet pressure too low. Inlet filter clogged.		
xWM39	15.6	7	28	30029	284	40285	6	14	6	39	79.6	0039	Drag indicator p2 ‡	Discharge pressure too high		
xWM40	15.7	7	28	30029	284	40285	7	14	7	39	79.7	0040	Overvoltage suppressor	Overvoltage at power supply.		
xWM41	14.0	7	28	30029	284	40285	8	15	0	39	78.0		free			
xWM42	14.1	7	28	30029	284	40285	9	15	1	39	78.1		free			
xWM43	14.2	7	28	30029	284	40285	10	15	2	39	78.2		free			
xWM44	14.3	7	28	30029	284	40285	11	15	3	39	78.3		free			
xWM45	14.4	7	28	30029	284	40285	12	15	4	39	78.4		free			
xWM46	14.5	7	28	30029	284	40285	13	15	5	39	78.5		free			
xWM47	14.6	7	28	30029	284	40285	14	15	6	39	78.6	0212	Electrical equipment h ‡	Electric components and installation: Maintenance interval has elapsed		
xWM48	14.7	7	28	30029	284	40285	15	15	7	39	78.7	0204	Motor bearings lube h ↑	Lubrication of motor bearing of drive motor: Maintenance interval has elapsed		

1. Blower ==>>> Master Controller

Table 1.2 : Warning Messages

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 2

* Summary of SC2 messages

JSON-RPC BDSnd	Profibus-net		Sigma Control 2 Modbus**				Modbus S5 Bit	DeviceNet EtherNet/IP		SAM Send/Receive		Message		Description / possible cause
	S7 DBX	S5 DW	Input Protocol address	Input Register	Holding Protocol address	Holding Register		Register	Bit	S5 DW	S7 DBX			
xWM49	17.0	8	29	30030	285	40286	0	16	0	40	81.0	0210	Annual maintenance due	Appears one year after last maintenance
xWM50	17.1	8	29	30030	285	40286	1	16	1	40	81.1	0300	SD card write error	Fault while writing data to SD card
xWM51	17.2	8	29	30030	285	40286	2	16	2	40	81.2	0050	Relubrication device	Relubrication device has been reporting an error
xWM52	17.3	8	29	30030	285	40286	3	16	3	40	81.3		free	
xWM53	17.4	8	29	30030	285	40286	4	16	4	40	81.4		free	
xWM54	17.5	8	29	30030	285	40286	5	16	5	40	81.5		free	
xWM55	17.6	8	29	30030	285	40286	6	16	6	40	81.6		free	
xWM56	17.7	8	29	30030	285	40286	7	16	7	40	81.7		free	
xWM57	16.0	8	29	30030	285	40286	8	17	0	40	80.0		free	
xWM58	16.1	8	29	30030	285	40286	9	17	1	40	80.1	0058	SC2 <=> SC2 communication error	Communication fault between SC2 controllers
xWM59	16.2	8	29	30030	285	40286	10	17	2	40	80.2		free	
xWM60	16.3	8	29	30030	285	40286	11	17	3	40	80.3		free	
xWM61	16.4	8	29	30030	285	40286	12	17	4	40	80.4		free	
xWM62	16.5	8	29	30030	285	40286	13	17	5	40	80.5	0024	FC service mode activated	FC in service mode
xWM63	16.6	8	29	30030	285	40286	14	17	6	40	80.6	0025	Test of cut-of paths required	STO ("Safe Torque off") channels must be tested
xWM64	16.7	8	29	30030	285	40286	15	17	7	40	80.7		free	

1. Blower ==>>> Master Controller

Table 1.2 : Warning Messages

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 2

* Summary of SC2 messages

JSON-RPC BDSnd	Profibus-net		Sigma Control 2 Modbus**				Modbus S5 Bit	DeviceNet EtherNet/IP		SAM Send/Receive		Message		Description / possible cause
	S7 DBX	S5 DW	Input Protocol address	Input Register	Holding Protocol address	Holding Register		Register	Bit	S5 DW	S7 DBX	ID	Text (SC2 Display, except *)	
xWM65	19.0	9	30	30031	286	40287	0	18	0	41	83.0		free	
xWM66	19.1	9	30	30031	286	40287	1	18	1	41	83.1		free	
xWM67	19.2	9	30	30031	286	40287	2	18	2	41	83.2		free	
xWM68	19.3	9	30	30031	286	40287	3	18	3	41	83.3	0098	T-Switch T3	
xWM69	19.4	9	30	30031	286	40287	4	18	4	41	83.4	0099	T-Switch T_S	
xWM70	19.5	9	30	30031	286	40287	5	18	5	41	83.5	0080	Vibration monitoring VSE ‡	Vibrations at block or motor too high
xWM71	19.6	9	30	30031	286	40287	6	18	6	41	83.6	0081	Vibration monitoring VSE †	Vibrations at block or motor high
xWM72	19.7	9	30	30031	286	40287	7	18	7	41	83.7	0082	Vibration monitoring sum signal ↑	Vibrations at block or motor high
xWM73	18.0	9	30	30031	286	40287	8	19	0	41	82.0	0014	External message 1	External binary message 1
xWM74	18.1	9	30	30031	286	40287	9	19	1	41	82.1	0074	External message 2	External binary message 2
xWM75	18.2	9	30	30031	286	40287	10	19	2	41	82.2	0075	External message 3	External binary message 3
xWM76	18.3	9	30	30031	286	40287	11	19	3	41	82.3	0092	p-Switch p1	Adjustable p1-pressure switch
xWM77	18.4	9	30	30031	286	40287	12	19	4	41	82.4	0093	p-Switch p2	Adjustable p2-pressure switch
xWM78	18.5	9	30	30031	286	40287	13	19	5	41	82.5	0094	T-Switch T1	Adjustable T1-temperature switch
xWM79	18.6	9	30	30031	286	40287	14	19	6	41	82.6	0095	T-Switch T2	Adjustable T2-temperature switch
xWM80	18.7	9	30	30031	286	40287	15	19	7	41	82.7	0096	Block speed switch	Adjustable block speed switch

1. Blower ==>>>> Master Controller

Table 1.3 : Operational messages

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 2

* Summary of SC2 messages

JSON-RPC	Sigma Control 2										SAM		Message		Description / possible cause
	Profibus-net		Modbus**				DeviceNet EtherNet/IP	Send/Receive							
	BDSnd	S7 DBX	S5 DW	Input address	Input Register	Holding address		Holding Register	Modbus S5 Bit	Register	Bit	S5 DW	S7 DBX	ID	
xOM1	21.0	10	31	30032	287	40288	0	20	0	42	85.0		free		
xOM2	21.1	10	31	30032	287	40288	1	20	1	42	85.1		free		
xOM3	21.2	10	31	30032	287	40288	2	20	2	42	85.2		free		
xOM4	21.3	10	31	30032	287	40288	3	20	3	42	85.3		free		
xOM5	21.4	10	31	30032	287	40288	4	20	4	42	85.4		free		
xOM6	21.5	10	31	30032	287	40288	5	20	5	42	85.5		free		
xOM7	21.6	10	31	30032	287	40288	6	20	6	42	85.6		free		
xOM8	21.7	10	31	30032	287	40288	7	20	7	42	85.7		free		
xOM9	20.0	10	31	30032	287	40288	8	21	0	42	84.0		free		
xOM10	20.1	10	31	30032	287	40288	9	21	1	42	84.1		free		
xOM11	20.2	10	31	30032	287	40288	10	21	2	42	84.2	0011	Waiting for cold start enabling	The optional heating has not yet heated the blower sufficiently	
xOM12	20.3	10	31	30032	287	40288	11	21	3	42	84.3	0012	Oil return started	The solenoid valve for the gear ventilation has opened	
xOM13	20.4	10	31	30032	287	40288	12	21	4	42	84.4	0013	Low oil, wait for oil return flow	Oil level too low, blower does not start as long as the oil level is too low and the oil return is active.	
xOM14	20.5	10	31	30032	287	40288	13	21	5	42	84.5		free		
xOM15	20.6	10	31	30032	287	40288	14	21	6	42	84.6		free		
xOM16	20.7	10	31	30032	287	40288	15	21	7	42	84.7		free		

SIGMA CONTROL 2 Process Map

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Table 1.3

1. Blower ==>>> Master Controller

Table 1.3 : Operational messages

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 2

* Summary of SC2 messages

JSON-RPC	Sigma Control 2										SAM		Message		Description / possible cause
	Profibus/-net		Modbus**				Modbus	DeviceNet		Send/Receive					
	BDSnd	S7 DBX	S5 DW	Input address	Input Register	Holding address		Holding Register	S5 Bit	Register	Bit	S5 DW	S7 DBX		
			Protocol address	Input Register	Protocol address	Holding Register		Register	Bit			ID	Text (SC2 Display, except *)		
xOM17	23.0	11	32	30033	288	40289	0	22	0	43	87.0	0073	External message 1	External binary message 1	
xOM18	23.1	11	32	30033	288	40289	1	22	1	43	87.1	0074	External message 2	External binary message 2	
xOM19	23.2	11	32	30033	288	40289	2	22	2	43	87.2	0075	External message 3	External binary message 3	
xOM20	23.3	11	32	30033	288	40289	3	22	3	43	87.3	0080	Vibration monitoring VSE ↑	Vibrations at block or motor high	
xOM21	23.4	11	32	30033	288	40289	4	22	4	43	87.4	0081	Vibration monitoring sum signal ↑	Vibrations at block or motor high	
xOM22	23.5	11	32	30033	288	40289	5	22	5	43	87.5		free		
xOM23	23.6	11	32	30033	288	40289	6	22	6	43	87.6		free		
xOM24	23.7	11	32	30033	288	40289	7	22	7	43	87.7		free		
xOM25	22.0	11	32	30033	288	40289	8	23	0	43	86.0		free		
xOM26	22.1	11	32	30033	288	40289	9	23	1	43	86.1		free		
xOM27	22.2	11	32	30033	288	40289	10	23	2	43	86.2	0027	Power OFF -> ON	It is necessary to switch power supply of SC2 off and on.	
xOM28	22.3	11	32	30033	288	40289	11	23	3	43	86.3		free		
xOM29	22.4	11	32	30033	288	40289	12	23	4	43	86.4		free		
xOM30	22.5	11	32	30033	288	40289	13	23	5	43	86.5	0092	p-Switch p1	Adjustable p1-pressure switch	
xOM31	22.6	11	32	30033	288	40289	14	23	6	43	86.6	0093	p-Switch p2	Adjustable p2-pressure switch	
xOM32	22.7	11	32	30033	288	40289	15	23	7	43	86.7	0094	T-Switch T1	Adjustable T1-temperature switch	

1. Blower ==>>>> Master Controller

Table 1.3 : Operational messages

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 2

* Summary of SC2 messages

JSON-RPC	Sigma Control 2									SAM		Message		Description / possible cause
	Profibus/net		Modbus**				Modbus	DeviceNet EtherNet/IP		Send/Receive				
			Input		Holding			Register	Register	Bit	Bit	S5	S7	
BDSnd	S7 DBX	S5 DW	Protocol address	Input Register	Protocol address	Holding Register	S5 Bit	Register	Bit	S5 DW	S7 DBX	ID	Text (SC2 Display, except *)	
xOM33	25.0	12	33	30034	289	40290	0	24	0	44	89.0	0095	T-Switch T2	Adjustable T2-temperature switch
xOM34	25.1	12	33	30034	289	40290	1	24	1	44	89.1	0096	Block speed switch	Adjustable block speed switch
xOM35	25.2	12	33	30034	289	40290	2	24	2	44	89.2	0097	T-Switch T3	
xOM36	25.3	12	33	30034	289	40290	3	24	3	44	89.3	0098	T-Switch T_S	
xOM37	25.4	12	33	30034	289	40290	4	24	4	44	89.4		free	
xOM38	25.5	12	33	30034	289	40290	5	24	5	44	89.5	0038	Drag indicator p1 ‡	Inlet pressure too low. Inlet filter clogged.
xOM39	25.6	12	33	30034	289	40290	6	24	6	44	89.6	0039	Drag indicator p2 ‡	Discharge pressure too high
xOM40	25.7	12	33	30034	289	40290	7	24	7	44	89.7		free	

SIGMA CONTROL 2 Process Map

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Table 1.4

1. Blower ==>>> Master Controller

Table 1.4 : Binary Signals

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 2

JSON-RPC	Sigma Control 2										SAM		Description	Signal	
	Profibus/-net		Modbus**				Modbus	DeviceNet		Send/Receive					
	BDSnd	S7 DBX	S5 DW	Input address	Input Register	Holding address		Holding Register	S5 Bit	EtherNet/IP	Register	Bit		S5 DW	S7 DBX
xBS0	27.0	13	34	30035	290	40291	0	26	0	45	91.0	Controller on (LED at key "I" on) + Remote on + Bus (Receive)	yes	no	
xBS1	27.1	13	34	30035	290	40291	1	26	1	45	91.1	Motor running	yes	no	
xBS2	27.2	13	34	30035	290	40291	2	26	2	45	91.2	Load run	yes	no	
xBS3	27.3	13	34	30035	290	40291	3	26	3	45	91.3	Idle	yes	no	
xBS4	27.4	13	34	30035	290	40291	4	26	4	45	91.4	Local mode (LED at key "Remote" off)	yes	no	
xBS5	27.5	13	34	30035	290	40291	5	26	5	45	91.5	Controller on (LED at key "I" on)	yes	no	
xBS6	27.6	13	34	30035	290	40291	6	26	6	45	91.6	Group alarm	yes	no	
xBS7	27.7	13	34	30035	290	40291	7	26	7	45	91.7	Group warning or group maintenance	yes	no	
xBS8	26.0	13	34	30035	290	40291	8	27	0	45	90.0				
xBS9	26.1	13	34	30035	290	40291	9	27	1	45	90.1	Blower ON (Motor is running or ready for operation)	yes	no	
xBS10	26.2	13	34	30035	290	40291	10	27	2	45	90.2	Aftercooler fan is running	yes	no	
xBS11	26.3	13	34	30035	290	40291	11	27	3	45	90.3	Sound enclosure fan is running	yes	no	
xBS12	26.4	13	34	30035	290	40291	12	27	4	45	90.4				
xBS13	26.5	13	34	30035	290	40291	13	27	5	45	90.5				
xBS14	26.6	13	34	30035	290	40291	14	27	6	45	90.6	Group maintenance (maintenance messages only)	yes	no	
xBS15	26.7	13	34	30035	290	40291	15	27	7	45	90.7	Group warning (warning messages only)	yes	no	

SIGMA CONTROL 2 Process Map

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Table 1.4

1. Blower ==>>> Master Controller

Table 1.4 : Binary Signals

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 2

JSON-RPC	Sigma Control 2										Description	Signal		
	Profibus/-net		Modbus**				DeviceNet EtherNet/IP	SAM						
	BDSnd	S7 DBX	S5 DW	Input address	Input Register	Holding address		Holding Register	Modbus S5 Bit	Register		Bit	S5 DW	S7 DBX
xBS16	29.0	14	35	30036	291	40292	0	28	0	46	93.0	Mode <i>blower ON</i> from "I" key only is active	yes	no
xBS17	29.1	14	35	30036	291	40292	1	28	1	46	93.1			
xBS18	29.2	14	35	30036	291	40292	2	28	2	46	93.2			
xBS19	29.3	14	35	30036	291	40292	3	28	3	46	93.3			
xBS20	29.4	14	35	30036	291	40292	4	28	4	46	93.4			
xBS21	29.5	14	35	30036	291	40292	5	28	5	46	93.5			
xBS22	29.6	14	35	30036	291	40292	6	28	6	46	93.6			
xBS23	29.7	14	35	30036	291	40292	7	28	7	46	93.7			
xBS24	28.0	14	35	30036	291	40292	8	29	0	46	92.0			
xBS25	28.1	14	35	30036	291	40292	9	29	1	46	92.1			
xBS26	28.2	14	35	30036	291	40292	10	29	2	46	92.2			
xBS27	28.3	14	35	30036	291	40292	11	29	3	46	92.3			
xBS28	28.4	14	35	30036	291	40292	12	29	4	46	92.4			
xBS29	28.5	14	35	30036	291	40292	13	29	5	46	92.5			
xBS30	28.6	14	35	30036	291	40292	14	29	6	46	92.6			
xBS31	28.7	14	35	30036	291	40292	15	29	7	46	92.7			

1. Blower ==>>> Master Controller

Table 1.4 : Binary Signals

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 2

JSON-RPC	Sigma Control 2											Description	Signal	
	Profibus/-net		Modbus**				Modbus	DeviceNet EtherNet/IP		SAM Send/Receive				
	BDSnd	S7 DBX	S5 DW	Input Protocol address	Input Register	Holding Protocol address		Holding Register	S5 Bit	Register	Bit		S5 DW	S7 DBX
xBS32	31.0	15	36	30037	292	40293	0	30	0	47	95.0			
xBS33	31.1	15	36	30037	292	40293	1	30	1	47	95.1			
xBS34	31.2	15	36	30037	292	40293	2	30	2	47	95.2			
xBS35	31.3	15	36	30037	292	40293	3	30	3	47	95.3			
xBS36	31.4	15	36	30037	292	40293	4	30	4	47	95.4	Mode <i>load-idle signal from external contact</i> is active	yes	no
xBS37	31.5	15	36	30037	292	40293	5	30	5	47	95.5			
xBS38	31.6	15	36	30037	292	40293	6	30	6	47	95.6	Mode <i>load-idle signal from bus master</i> is active	yes	no
xBS39	31.7	15	36	30037	292	40293	7	30	7	47	95.7			
xBS40	30.0	15	36	30037	292	40293	8	31	0	47	94.0			
xBS41	30.1	15	36	30037	292	40293	9	31	1	47	94.1			
xBS42	30.2	15	36	30037	292	40293	10	31	2	47	94.2			
xBS43	30.3	15	36	30037	292	40293	11	31	3	47	94.3			
xBS44	30.4	15	36	30037	292	40293	12	31	4	47	94.4			
xBS45	30.5	15	36	30037	292	40293	13	31	5	47	94.5			
xBS46	30.6	15	36	30037	292	40293	14	31	6	47	94.6			
xBS47	30.7	15	36	30037	292	40293	15	31	7	47	94.7			

SIGMA CONTROL 2 Process Map

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Table 1.4

1. Blower ==>>> Master Controller

Table 1.4 : Binary Signals

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 2

JSON-RPC	Sigma Control 2											Description	Signal	
	Profibus/-net		Modbus**				Modbus	DeviceNet EtherNet/IP		SAM Send/Receive				
	BDSnd	S7 DBX	S5 DW	Input address	Input Register	Holding address		Holding Register	S5 Bit	Register	Bit		S5 DW	S7 DBX
xBS48	33.0	16	37	30038	293	40294	0	32	0	48	97.0			
xBS49	33.1	16	37	30038	293	40294	1	32	1	48	97.1			
xBS50	33.2	16	37	30038	293	40294	2	32	2	48	97.2			
xBS51	33.3	16	37	30038	293	40294	3	32	3	48	97.3	Automatic restart after power on	activated	deactivated
xBS52	33.4	16	37	30038	293	40294	4	32	4	48	97.4			
xBS53	33.5	16	37	30038	293	40294	5	32	5	48	97.5			
xBS54	33.6	16	37	30038	293	40294	6	32	6	48	97.6			
xBS55	33.7	16	37	30038	293	40294	7	32	7	48	97.7	Message acknowledgement by bus master	activated	deactivated
xBS56	32.0	16	37	30038	293	40294	8	33	0	48	96.0			
xBS57	32.1	16	37	30038	293	40294	9	33	1	48	96.1			
xBS58	32.2	16	37	30038	293	40294	10	33	2	48	96.2			
xBS59	32.3	16	37	30038	293	40294	11	33	3	48	96.3			
xBS60	32.4	16	37	30038	293	40294	12	33	4	48	96.4			
xBS61	32.5	16	37	30038	293	40294	13	33	5	48	96.5			
xBS62	32.6	16	37	30038	293	40294	14	33	6	48	96.6			
xBS63	32.7	16	37	30038	293	40294	15	33	7	48	96.7			

SIGMA CONTROL 2 Process Map

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Table 2 Process

2. Master Controller ==>>> Blower

Table 2: Process data

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 2

Sigma Control 2						SAM		Description	Unit	Format	
JSON-RPC	Profibus/Profinet		Modbus**		DeviceNet EtherNet/IP	Send/Receive					
BDRcv	S7 DBW	S5 DW	Protocol address	Holding Register	Register MSB/LSB	S5 DW	S7 DBW				
	0	0	0	40001	1/0	0	0	Receive bit xBS15 - xBS0	Table 2.1	binary	BOOL
	2	1	1	40002	3/2	1	2	Receive bit xBS31 - xBS16	Table 2.1	binary	BOOL
	4	2	2	40003	5/4	2	4	Receive bit xBS47 - xBS32	Table 2.1	binary	BOOL
	6	3	3	40004	7/6	3	6	Receive bit xBS63 - xBS48	Table 2.1	binary	BOOL
iDV0	8	4	4	40005	9/8	4	8	Setpoint pressure	<i>if xBS68=1</i>	mbar	INT
iDV1	10	5	5	40006	11/10	5	10				
iDV2	12	6	6	40007	13/12	6	12				
iDV3	14	7	7	40008	15/14	7	14				
iDV4	16	8	8	40009	17/16	8	16	Set block speed	<i>if xBS48=1</i>	1/min	INT
iDV5	18	9	9	40010	19/18	9	18	Set value for control of external variable	<i>if xBS49=1</i>	%	INT
iDV6	20	10	10	40011	21/20	10	20				
iDV7	22	11	11	40012	23/22	11	22				
iDV8	24	12	12	40013	25/24	12	24				
iDV9	26	13	13	40014	27/26	13	26	Actual air receiver system pressure	<i>if xBS77=1</i>	mbar	INT
iDV10	28	14	14	40015	29/28	14	28				
iDV11	30	15	15	40016	31/30	15	30				
iDV12	32	16	16	40017	33/32	16	32				
	34	17	17	40018	35/34	17	34	Receive bit xBS79 - xBS64	Table 2.1	binary	BOOL
	36	18	18	40019	37/36	18	36	Receive bit xBS95 - xBS80	Table 2.1	binary	BOOL
iMasterMsgHeader	38	19	19	40020	39/38	19	38	Message header master			WORD
iDV13	40	20	20	40021	41/40	20	40				
iDV14	42	21	21	40022	43/42	21	42				
iDV15	44	22	22	40023	45/44	22	44				
iDV16	46	23	23	40024	47/46	23	46				
iDV17	48	24	24	40025	49/48	24	48				
iDV18	50	25	25	40026	51/50	25	50				
iDV19	52	26	26	40027	53/52	26	52				
iDV20	54	27	27	40028	55/54	27	54				
iDV21	56	28	28	40029	57/56	28	56				
iDV22	58	29	29	40030	59/58	29	58				
iDV23	60	30	30	40031	61/60	30	60				
iMasterMsgFooter	62	31	31	40032	63/62	31	62	Message footer master			WORD

SIGMA CONTROL 2 Process Map

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Table 2 Buspar

2. Master Controller ==>>>> Blower

Table 2: Busparameter, Modbus only

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 2

* not for Modbus TCP

Sigma Control 2						SAM			Description	Example		
JSON-RPC	Profibus/Profinet		Modbus **			DeviceNet EtherNet/IP	Send/Receive			HEX	Value	Result
BDSnd	S7 DBB	S5 DW	Protocol address	Holding Register	Bit	Register MSB/LSB	S5 DW	S7 DBB				
%	%	%	512	40513		%	%	%	Modbus: Slaveaddress	64	100	Slave: 100
%	%	%	513	40514		%	%	%	Modbus: Priority, stop bits, baudrate *	16		19200/even/1stop
									Even parity, 1 stop bit (default)			
									Odd parity, 1 stop bit			
									No parity, 2 stop bits			
									No parity, 1 stop bit			
			513	40514	0				0 1 0 1			
			513	40514	1				0 0 1 1			
									1200bit/s			
									2400bit/s			
									4800bit/s			
									9600bit/s			
									19200bit/s (default)			
									38400bit/s			
									57600bit/s			
									76800bit/s			
			513	40514	2				0 1 0 1 0 1 0 1 0			
			513	40514	3				0 0 1 1 0 0 1 1 0			
			513	40514	4				0 0 0 0 1 1 1 1 0			
			513	40514	5				0 0 0 0 0 0 0 0 1			
			513	40514	6				reserved, set to 0			
			513	40514	7				reserved, set to 0			
%	%	%	514	40515		%	%	%	Modbus: Modus: 0=RTU, 1=ASCII *	0	RTU	Modus: RTU
%	%	%	515	40516		%	%	%	Modbus: Timeout in ms	7D0	2000	Timeout: 2000ms

2. Master Controller ==>>>> Blower

Table 2.1: Receive bit assignment

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 2

JSON-RPC	Sigma Control 2					DeviceNet EtherNet/IP		SAM Send/Receive		Description	Signal		
	Profibus/Profinet		Modbus**		Modbus	Register	Bit	S5 DW	S7 DBX		"1"	"0"	Kind
BDRcv	S7 DBX	S5 DW	Protocol address	Holding Register	S5 Bit	Register	Bit	S5 DW	S7 DBX				
xBS0	1.0	0	0	40001	0	0	0	0	1.0	SAM connection	yes	no	Duration
xBS1	1.1	0	0	40001	1	0	1	0	1.1	reserved			
xBS2	1.2	0	0	40001	2	0	2	0	1.2	Enable <i>Continue drive motor running</i> by bus master	yes (continue run with xBS3)	local	Duration
xBS3	1.3	0	0	40001	3	0	3	0	1.3	Continue drive motor running (<i>if xBS2=1</i>)	continue running	no bearing	Duration
xBS4	1.4	0	0	40001	4	0	4	0	1.4				
xBS5	1.5	0	0	40001	5	0	5	0	1.5				
xBS6	1.6	0	0	40001	6	0	6	0	1.6				
xBS7	1.7	0	0	40001	7	0	7	0	1.7				
xBS8	0.0	0	0	40001	8	1	0	0	0.0				
xBS9	0.1	0	0	40001	9	1	1	0	0.1				
xBS10	0.2	0	0	40001	10	1	2	0	0.2				
xBS11	0.3	0	0	40001	11	1	3	0	0.3				
xBS12	0.4	0	0	40001	12	1	4	0	0.4				
xBS13	0.5	0	0	40001	13	1	5	0	0.5				
xBS14	0.6	0	0	40001	14	1	6	0	0.6				
xBS15	0.7	0	0	40001	15	1	7	0	0.7				

2. Master Controller ==>>>> Blower

Table 2.1: Receive bit assignment

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 2

JSON-RPC	Sigma Control 2					DeviceNet EtherNet/IP		SAM Send/Receive		Description	Signal		
	Profibus/Profinet		Modbus**		Modbus	Register	Bit	S5 DW	S7 DBX		"1"	"0"	Kind
BDRcv	S7 DBX	S5 DW	Protocol address	Holding Register	S5 Bit	Register	Bit	S5 DW	S7 DBX				
xBS16	3.0	1	1	40002	0	2	0	1	3.0				
xBS17	3.1	1	1	40002	1	2	1	1	3.1				
xBS18	3.2	1	1	40002	2	2	2	1	3.2				
xBS19	3.3	1	1	40002	3	2	3	1	3.3				
xBS20	3.4	1	1	40002	4	2	4	1	3.4				
xBS21	3.5	1	1	40002	5	2	5	1	3.5				
xBS22	3.6	1	1	40002	6	2	6	1	3.6				
xBS23	3.7	1	1	40002	7	2	7	1	3.7				
xBS24	2.0	1	1	40002	8	3	0	1	2.0				
xBS25	2.1	1	1	40002	9	3	1	1	2.1				
xBS26	2.2	1	1	40002	10	3	2	1	2.2				
xBS27	2.3	1	1	40002	11	3	3	1	2.3				
xBS28	2.4	1	1	40002	12	3	4	1	2.4				
xBS29	2.5	1	1	40002	13	3	5	1	2.5				
xBS30	2.6	1	1	40002	14	3	6	1	2.6	Enable <i>Change set value for load control of external value</i> by bus master	yes (Change with xBS31)	local	Duration
xBS31	2.7	1	1	40002	15	3	7	1	2.7	Set value for load control (<i>if xBS30=1</i>)	xB	xA	Duration

2. Master Controller ==>>>> Blower

Table 2.1: Receive bit assignment

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 2

JSON-RPC	Sigma Control 2				Modbus	DeviceNet		SAM		Description	Signal		
	Profibus/Profinet		Modbus**			EtherNet/IP		Send/Receive			"1"	"0"	Kind
BDRcv	S7 DBX	S5 DW	Protocol address	Holding Register	S5 Bit	Register	Bit	S5 DW	S7 DBX				
xBS32	5.0	2	2	40003	0	4	0	2	5.0				
xBS33	5.1	2	2	40003	1	4	1	2	5.1				
xBS34	5.2	2	2	40003	2	4	2	2	5.2				
xBS35	5.3	2	2	40003	3	4	3	2	5.3				
xBS36	5.4	2	2	40003	4	4	4	2	5.4				
xBS37	5.5	2	2	40003	5	4	5	2	5.5				
xBS38	5.6	2	2	40003	6	4	6	2	5.6	Mode <i>load-idle</i> signal from bus master	yes (load/no load via xBS39)	local	duration
xBS39	5.7	2	2	40003	7	4	7	2	5.7	<i>Load run</i> by bus master (if xBS38 = 1)	load run	no load run	duration
xBS40	4.0	2	2	40003	8	5	0	2	4.0				
xBS41	4.1	2	2	40003	9	5	1	2	4.1				
xBS42	4.2	2	2	40003	10	5	2	2	4.2				
xBS43	4.3	2	2	40003	11	5	3	2	4.3				
xBS44	4.4	2	2	40003	12	5	4	2	4.4				
xBS45	4.5	2	2	40003	13	5	5	2	4.5				
xBS46	4.6	2	2	40003	14	5	6	2	4.6				
xBS47	4.7	2	2	40003	15	5	7	2	4.7				

SIGMA CONTROL 2 Process Map

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Table 2.1

2. Master Controller ==>>> Blower

Table 2.1: Receive bit assignment

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 2

JSON-RPC	Sigma Control 2					DeviceNet EtherNet/IP			SAM Send/Receive		Description	Signal		
	Profibus/Profinet		Modbus**		Modbus S5 Bit	Register	Bit	S5 DW	S7 DBX	"1"		"0"	Kind	
BDRcv	S7 DBX	S5 DW	Protocol address	Holding Register						S5 Bit	Register	Bit	S5 DW	S7 DBX
xBS48	7.0	3	3	40004	0	6	0	3	7.0	Set block speed via	bus master (iDV4)	local	duration	
xBS49	7.1	3	3	40004	1	6	1	3	7.1	Set value for control of external variable via	bus master (iDV5)	local	duration	
xBS50	7.2	3	3	40004	2	6	2	3	7.2					
xBS51	7.3	3	3	40004	3	6	3	3	7.3					
xBS52	7.4	3	3	40004	4	6	4	3	7.4					
xBS53	7.5	3	3	40004	5	6	5	3	7.5					
xBS54	7.6	3	3	40004	6	6	6	3	7.6	Enable message acknowledgement by bus master	yes (ack. via xBS55)	local	duration	
xBS55	7.7	3	3	40004	7	6	7	3	7.7	Acknowledge signal for alarms and warnings (<i>if xBS54=1</i>)	L→H: acknowledge		flank	
xBS56	6.0	3	3	40004	8	7	0	3	6.0	Enable blower autostart by bus master	yes (set via xBS57)	local	duration	
xBS57	6.1	3	3	40004	9	7	1	3	6.1	Blower restart (if xBS56=1)	yes	no	duration	
xBS58	6.2	3	3	40004	10	7	2	3	6.2					
xBS59	6.3	3	3	40004	11	7	3	3	6.3					
xBS60	6.4	3	3	40004	12	7	4	3	6.4					
xBS61	6.5	3	3	40004	13	7	5	3	6.5					
xBS62	6.6	3	3	40004	14	7	6	3	6.6					
xBS63	6.7	3	3	40004	15	7	7	3	6.7					

SIGMA CONTROL 2 Process Map

≥ blow_2.5.2

Table 2.1

2. Master Controller ==>>> Blower

Table 2.1: Receive bit assignment

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 2

JSON-RPC	Sigma Control 2					SAM		Description	Signal				
	Profibus/Profinet		Modbus**		Modbus S5 Bit	DeviceNet EtherNet/IP			Send/Receive		"1"	"0"	Kind
BDRcv	S7 DBX	S5 DW	Protocol address	Holding Register		Register	Bit	S5 DW	S7 DBX				
xBS64	35.0	17	17	40018	0	34	0	17	35.0				
xBS65	35.1	17	17	40018	1	34	1	17	35.1				
xBS66	35.2	17	17	40018	2	34	2	17	35.2				
xBS67	35.3	17	17	40018	3	34	3	17	35.3				
xBS68	35.4	17	17	40018	4	34	4	17	35.4	Set pressure via	bus master (iDV0)	local	duration
xBS69	35.5	17	17	40018	5	34	5	17	35.5				
xBS70	35.6	17	17	40018	6	34	6	17	35.6				
xBS71	35.7	17	17	40018	7	34	7	17	35.7				
xBS72	34.0	17	17	40018	8	35	0	17	34.0				
xBS73	34.1	17	17	40018	9	35	1	17	34.1				
xBS74	34.2	17	17	40018	10	35	2	17	34.2				
xBS75	34.3	17	17	40018	11	35	3	17	34.3				
xBS76	34.4	17	17	40018	12	35	4	17	34.4				
xBS77	34.5	17	17	40018	13	35	5	17	34.5	Get actual air receiver system pressure via	bus master (iDV9)	local	duration
xBS78	34.6	17	17	40018	14	35	6	17	34.6				
xBS79	34.7	17	17	40018	15	35	7	17	34.7				

SIGMA CONTROL 2 Process Map

≥ blow_2.5.2

Table 2.1

2. Master Controller ==>>>> Blower

Table 2.1: Receive bit assignment

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 2

JSON-RPC	Sigma Control 2					SAM		Description	Signal		
	Profibus/Profinet	Modbus**		Modbus S5 Bit	DeviceNet EtherNet/IP		Send/Receive		"1"	"0"	Kind
BDRcv	S7 DBX	S5 DW	Protocol address	Holding Register	Register	Bit	S5 DW	S7 DBX			
xBS80	37.0	18	18	40019	0	36	0	18	37.0		
xBS81	37.1	18	18	40019	1	36	1	18	37.1		
xBS82	37.2	18	18	40019	2	36	2	18	37.2		
xBS83	37.3	18	18	40019	3	36	3	18	37.3		
xBS84	37.4	18	18	40019	4	36	4	18	37.4		
xBS85	37.5	18	18	40019	5	36	5	18	37.5		
xBS86	37.6	18	18	40019	6	36	6	18	37.6		
xBS87	37.7	18	18	40019	7	36	7	18	37.7		
xBS88	36.0	18	18	40019	8	37	0	18	36.0		
xBS89	36.1	18	18	40019	9	37	1	18	36.1		
xBS90	36.2	18	18	40019	10	37	2	18	36.2		
xBS91	36.3	18	18	40019	11	37	3	18	36.3		
xBS92	36.4	18	18	40019	12	37	4	18	36.4		
xBS93	36.5	18	18	40019	13	37	5	18	36.5		
xBS94	36.6	18	18	40019	14	37	6	18	36.6		
xBS95	36.7	18	18	40019	15	37	7	18	36.7		